SFE316D GROUP PROJECT

Smart Waste Collection Management System (SWCMS)

2026 First Semester

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Instructions

- ✓ This is a group project.
- ✓ All the groups must have four (5) members.
- ✓ Each group approach must be unique; therefore, you are not expected to submit the same thing.
- ✓ Handwritten work will not be accepted.
- ✓ Submit a hard copy on 16 April 2026 before 13H30 at office 20-104.
- ✓ Late submissions will not be accepted.

BUSINESS PROFILING

Background and Scenario:
Many cities face difficulties managing solid waste collection efficiently. Garbage trucks follow fixed routes and schedules regardless of actual waste levels, causing overflowing bins in some areas and wasted fuel/time in others. Citizens also lack real-time information on waste pickup schedules or overflow alerts.

The city council seeks to develop a **Smart Waste Collection Management System (SWCMS)** to optimize garbage collection using smart IoT-enabled bins and route optimization.

1.1 Business Need and Objectives

1.1.1 System Vision Document

Field	Details
Project	Smart Waste Collection Management System (SWCMS)
Subsystem Focus	Bin Monitoring via IoT Sensors

Problem Description

Many municipalities currently operate garbage collection using fixed schedules and predetermined routes. This approach creates several critical inefficiencies that negatively affect public health, urban aesthetics, and municipal budgets.

- Waste bins in high-traffic areas overflow before scheduled collection, causing health hazards and environmental pollution.
- Trucks are dispatched to bins that are nearly empty, wasting fuel, driver time, and operational budget.
- There is no real-time visibility into bin fill levels across the city.
- Citizens have no way to report or track overflow situations effectively.
- City supervisors cannot make data-driven decisions about resource allocation.

System Capabilities

#	Capability
1	IoT ultrasonic sensors installed in each bin continuously measure fill levels in real time.
2	Each smart bin transmits fill-level data wirelessly to a central monitoring server.
3	The system generates automated alerts when a bin reaches a defined threshold (e.g., 80% full).
4	A central dashboard displays live bin status across all city zones on a map.
5	Historical fill-level data is logged for analysis and future route planning.
6	The system flags malfunctioning sensors and sends maintenance notifications.
7	Integration with the Route Optimization subsystem to trigger dynamic collection scheduling.

Business Benefits

Benefit	Description
Cost Reduction	Eliminates unnecessary truck dispatches to low-fill bins, reducing fuel and labour costs.
Improved Public Health	Prevents bin overflow, reducing exposure to disease-causing waste.
Operational Efficiency	Collection teams respond only when and where needed.
Data-Driven Decisions	Management gains insight into waste generation patterns per zone.

Environmental Impact	Reduced vehicle emissions from optimized collection runs.
Scalability	System can expand to monitor more bins as the city grows.

1.2 Business Analysis and Modeling

1.2.1 System Subsystems / Components

The SWCMS is divided into the following five (5) subsystems:

1. Bin Monitoring Subsystem (Selected Focus)
Uses IoT ultrasonic sensors embedded in smart bins to continuously measure and transmit real-time fill levels to the central server. Triggers alerts when thresholds are exceeded.

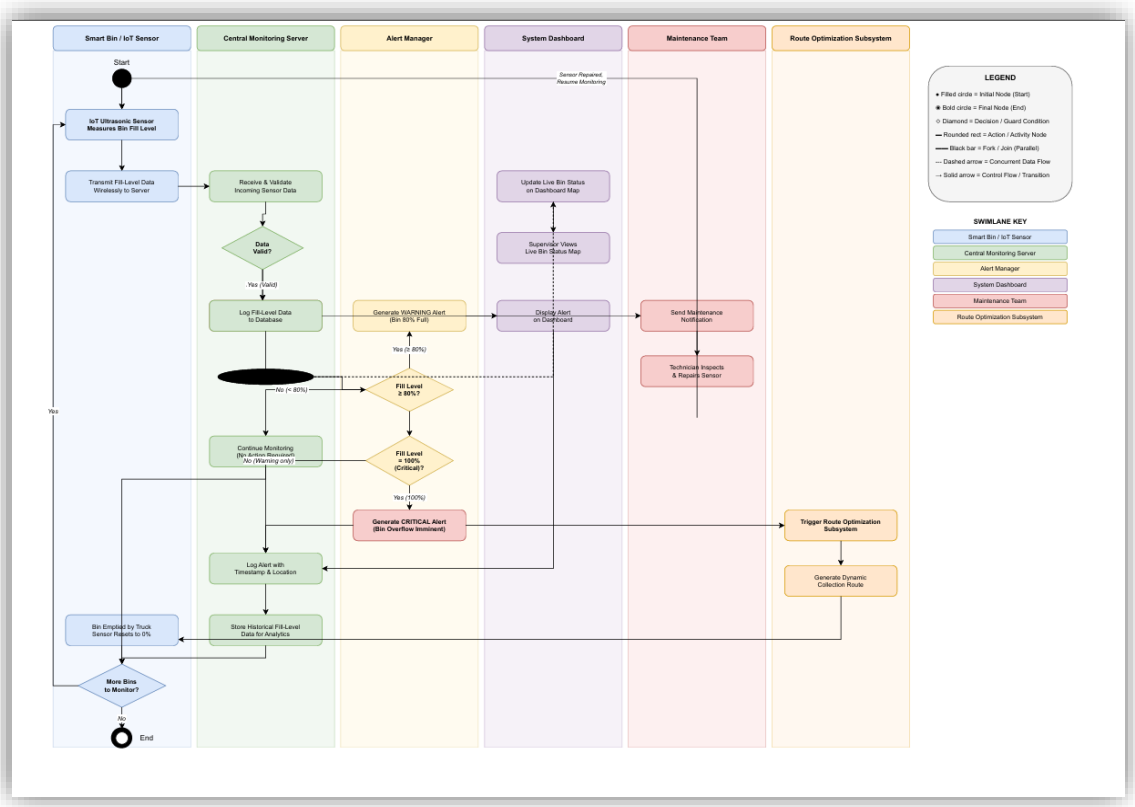
2. Route Optimization Subsystem
Receives fill-level alerts from the Bin Monitoring subsystem and dynamically generates the most efficient collection routes for garbage trucks, replacing fixed schedules.

3. Fleet & Driver Management Subsystem
Manages garbage truck assignments, driver schedules, GPS tracking of vehicles, and delivery of optimized routes to drivers via mobile devices.

4. Citizen Notification Portal
A web/mobile interface allowing residents to view bin collection schedules, report overflowing bins, and receive alerts about delays or service disruptions in their area.

5. Reporting & Analytics Subsystem
Aggregates data from all subsystems to generate operational reports, waste trend analyses, and performance dashboards for city management and decision-makers.

1.2.2 Workflow – Bin Monitoring Subsystem



2. Requirements Modelling

2.1 Use Case Modelling

2.1.1 User Stories for the users of our Selected Subsystem

a) **IoT Sensor** as an **IoT Sensor**, I want to continuously measure the bin fill level and transmit the data wirelessly to the central server so that the system always has up-to-date bin status information.

b) **City Supervisor**, I want to view a live map dashboard showing the fill level of all bins across the city so that I can monitor waste levels and make informed dispatch decisions.

c) **City Supervisor**, I want to receive automated alerts when a bin reaches 80% or 100% capacity so that I can act before overflow occurs.

d) **Maintenance Technician,** as a **Maintenance Technician**, I want to receive a notification when a bin sensor is flagged as malfunctioning so that I can inspect and repair it before it causes data gaps.

e) **Citizen** As a **Citizen**, I want to report an overflowing bin through a portal so that the city can respond quickly to public health hazards in my area.

f) **City Planner / Management** as a **City Planner**, I want to access historical fill-level data and waste trend reports so that I can make data-driven decisions about bin placement and collection frequency.

2.1.2 Even Decomposition

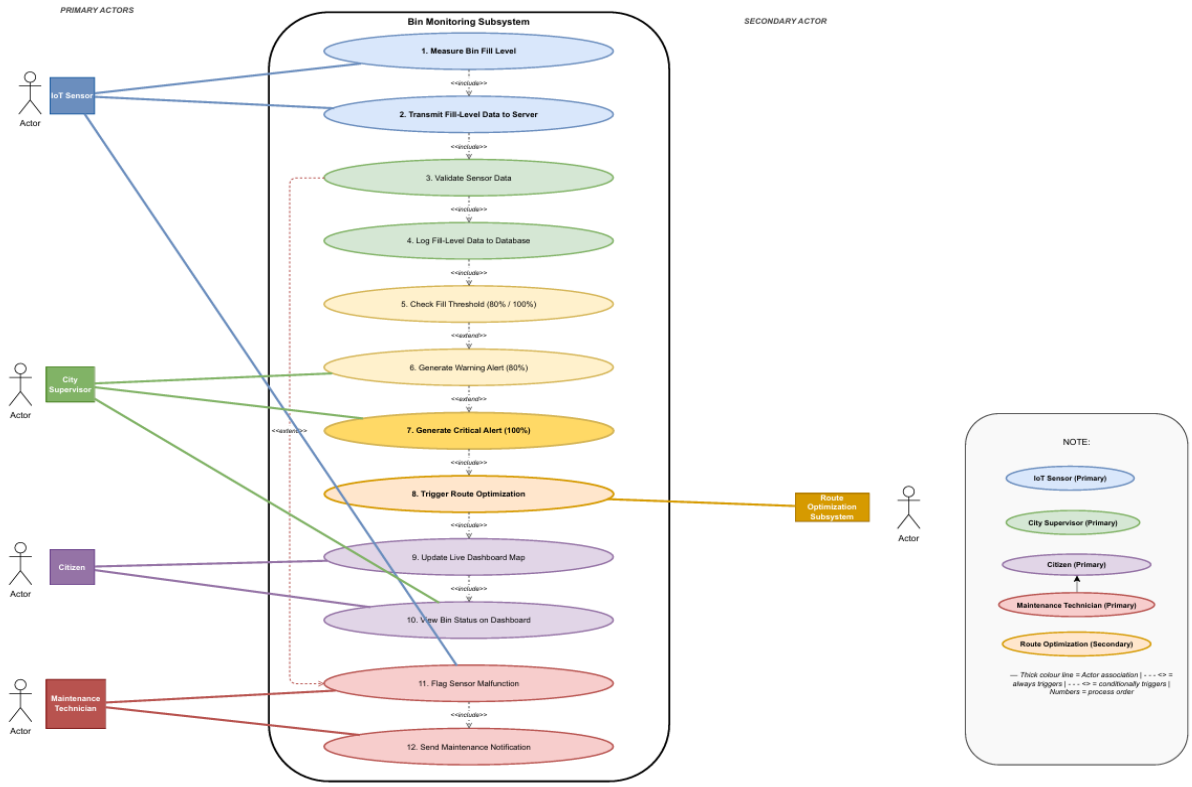
#	Event Type	Event	Use Case Identified
1	External	Sensor reads bin fill level	**Measure Bin Fill Level**
2	External	Sensor transmits data to server	**Transmit Fill-Level Data to Server**
3	External	Server receives sensor data	**Validate Sensor Data**
4	Temporal	System checks data at each cycle	**Log Fill-Level Data to Database**
5	State change	Fill level reaches 80%	**Generate Warning Alert**
6	State change	Fill level reaches 100%	**Generate Critical Alert**
7	Temporal	After every data log	**Update Live Dashboard Map**
8	External	Supervisor opens dashboard	**View Bin Status on Dashboard**

9	State change	Sensor returns invalid/no data	**Flag Sensor Malfunction**
10	External	System detects malfunction	**Send Maintenance Notification**
11	External	Critical alert is triggered	**Trigger Route Optimization**

2.1.3 CRUD Technique

Entity	Create	Read	Update	Delete
Bin	Register New Bin	View Bin Details / Location	Update Bin Status (fill %, GPS)	Decommission Bin
Sensor	Install / Register Sensor	View Sensor Health Status	Update Sensor Calibration	Remove Faulty Sensor
Alert	Generate Warning / Critical Alert	View Active Alerts	Acknowledge / Resolve Alert	Purge Resolved Alerts
Fill-Level Record	Log Fill-Level Data	View Historical Fill Data	Correct Erroneous Reading	Archive Old Records
Maintenance Report	Create Maintenance Request	View Maintenance History	Update Repair Status	Delete Closed Report

2.1.4 USE CASE DIAGRAM



3. DETAILED REQUIREMENTS MODELLING

3.1 Fully Developed Use Case Descriptions

Use Case 5: Check Fill Threshold

Field	Details
Use Case Name	Check Fill Threshold
Use Case ID	UC5
Actor(s)	IoT Sensor (primary), City Supervisor (primary)
Trigger	Sensor transmits a fill-level reading to the central server
Preconditions	Sensor is active; server is online; fill-level data has been received and validated
Main Flow	1. Sensor reads and transmits fill level to server. 2. Server logs the reading. 3. Alert Manager compares fill level to thresholds. 4. If < 80%, no action — monitoring continues. 5. If ≥ 80%, Warning Alert is generated. 6. If = 100%, Critical Alert is generated. 7. Alert is

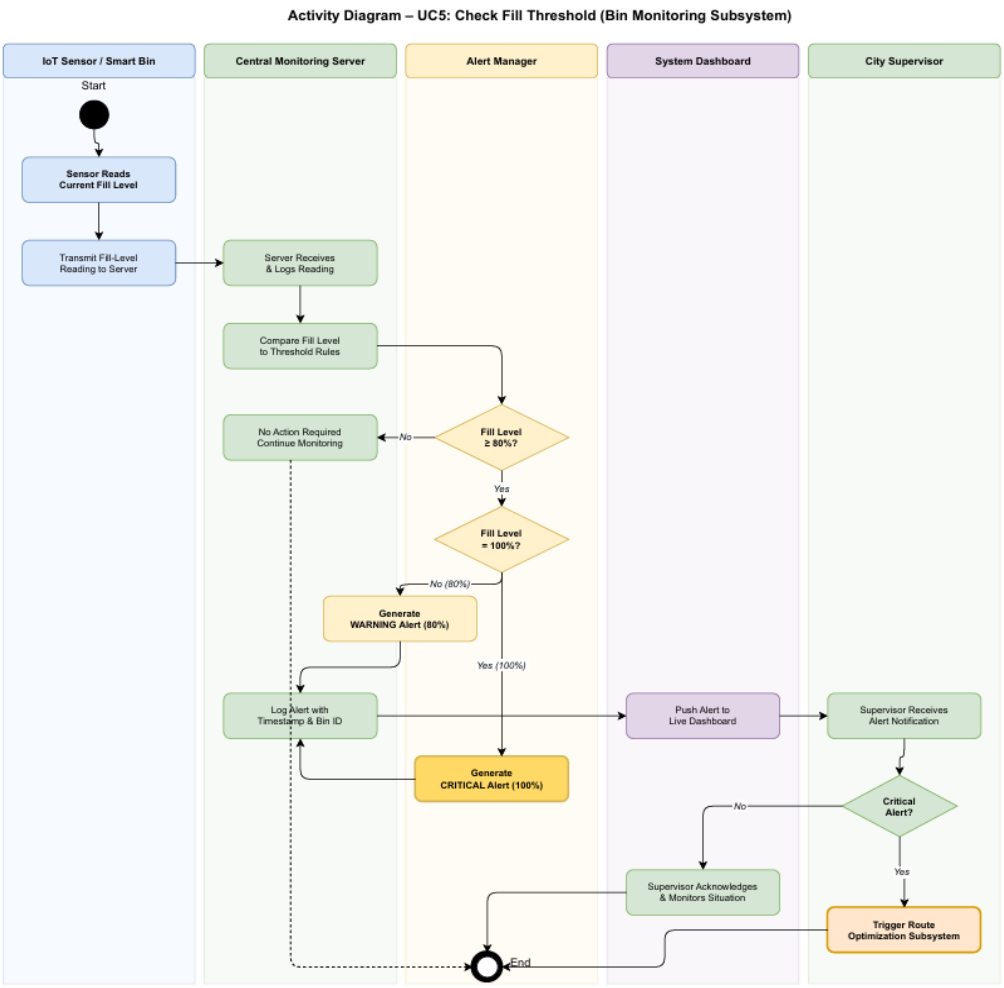
	logged with timestamp and bin ID. 8. Dashboard is updated and supervisor is notified. 9. If critical, Route Optimization is triggered.
Postconditions	Alert is logged; supervisor is notified; route optimization triggered if critical
Alternative Flow	If fill level drops below 80% before collection, warning alert is cleared and monitoring resumes

Use Case 11: Flag Sensor Malfunction

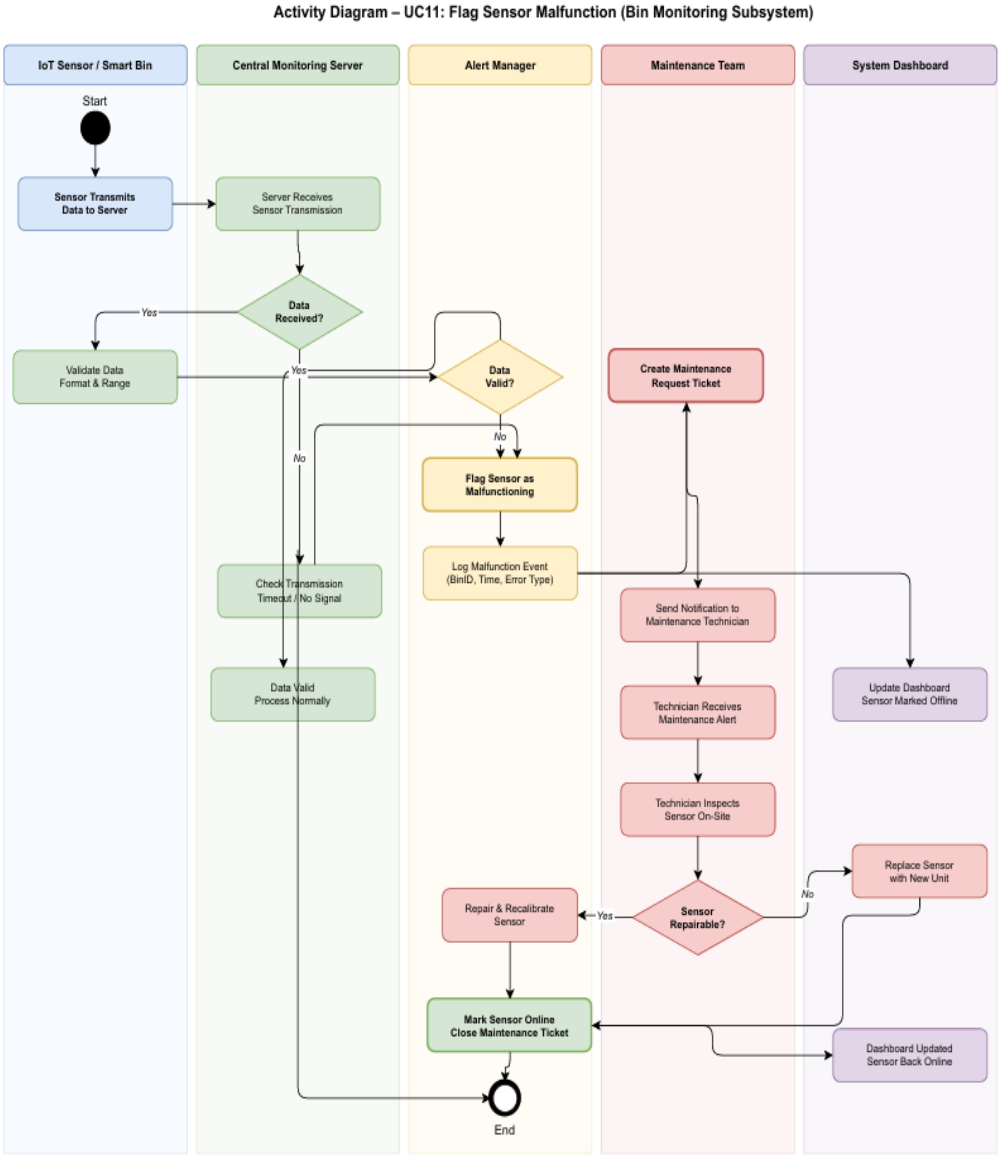
Field	**Details**
Use Case Name	Flag Sensor Malfunction
Use Case ID	UC11
Actor(s)	IoT Sensor (primary), Maintenance Technician (primary), System Dashboard
Trigger	Server detects invalid data or no signal from a sensor
Preconditions	Sensor has been transmitting; server is online and monitoring transmissions
Main Flow	1. Sensor transmits data to server. 2. Server checks data integrity. 3. If data is invalid or signal is lost after retries, Alert Manager flags sensor as malfunctioning. 4. Malfunction event is logged. 5. Maintenance ticket is created. 6. Dashboard marks sensor offline. 7. Maintenance notification sent to technician. 8. Technician inspects and repairs or replaces sensor. 9. Malfunction flag is cleared; sensor marked back online.
Postconditions	Sensor is repaired/replaced; malfunction flag cleared; dashboard updated; monitoring resumes
Alternative Flow	If data error resolves on retry (within 3 attempts), sensor returns to Active state without flagging

3.2 Activity Diagrams

Use Case 5: Check Fill Threshold

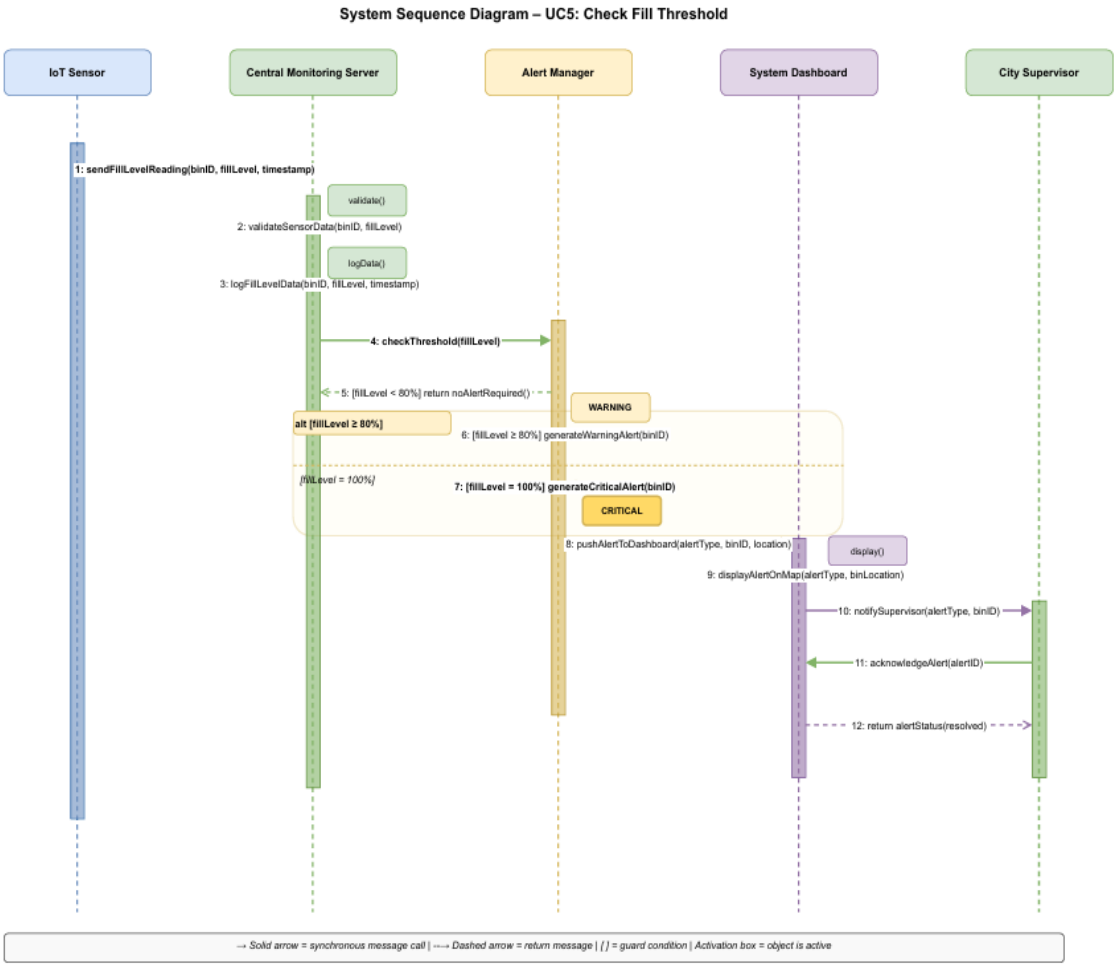


Use Case 11: Flag Sensor Malfunction

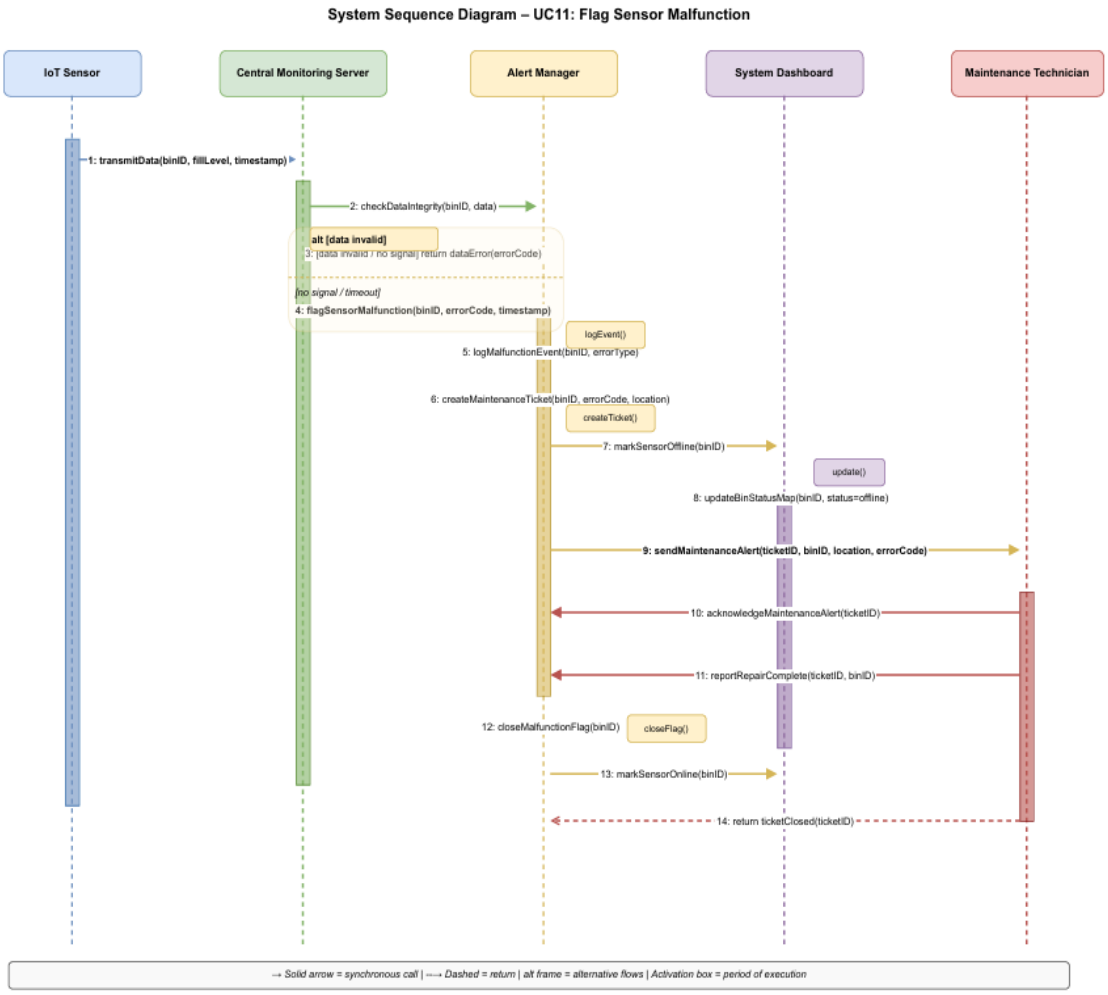


3.3 System Sequence Diagram

Use Case 5: Check Fill Threshold

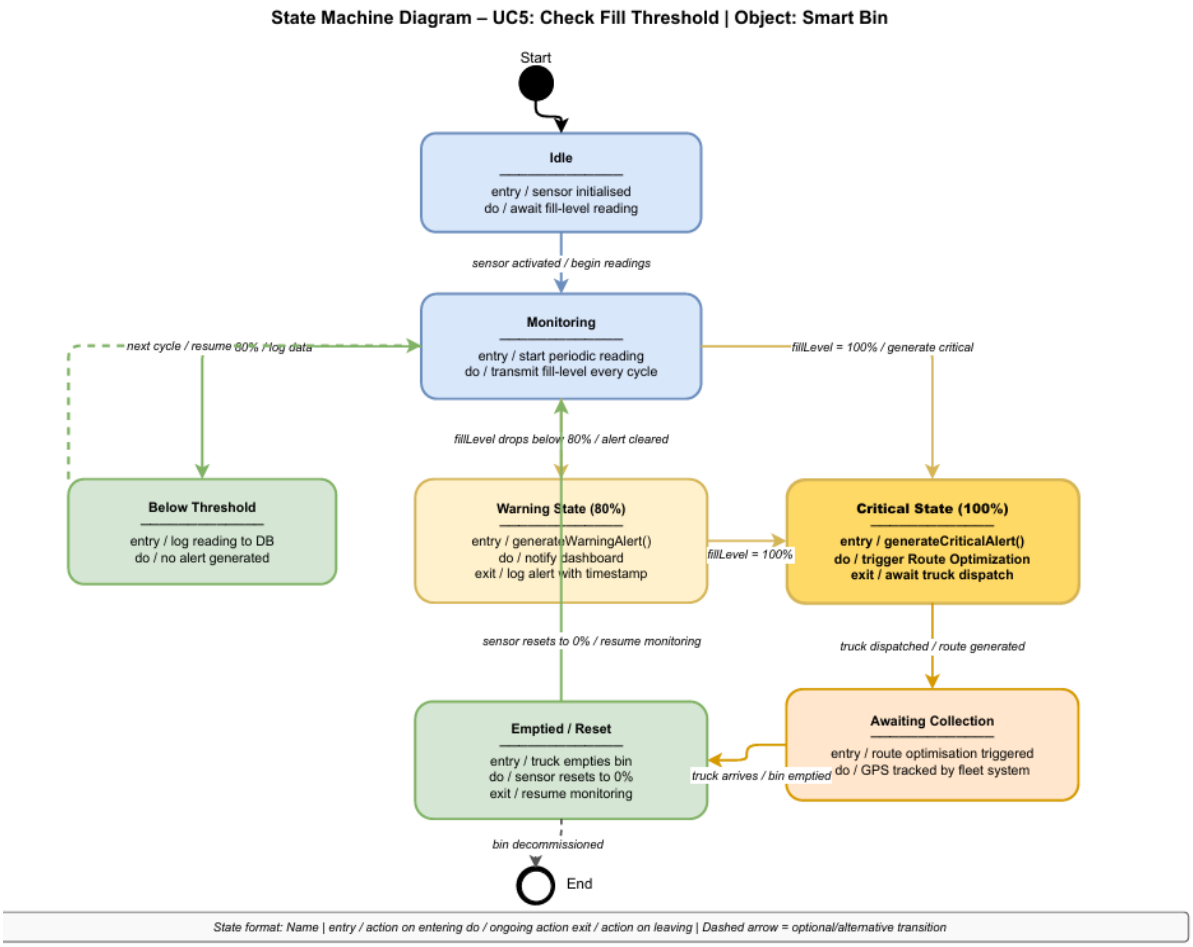


Use Case 11: Flag Sensor Malfunction

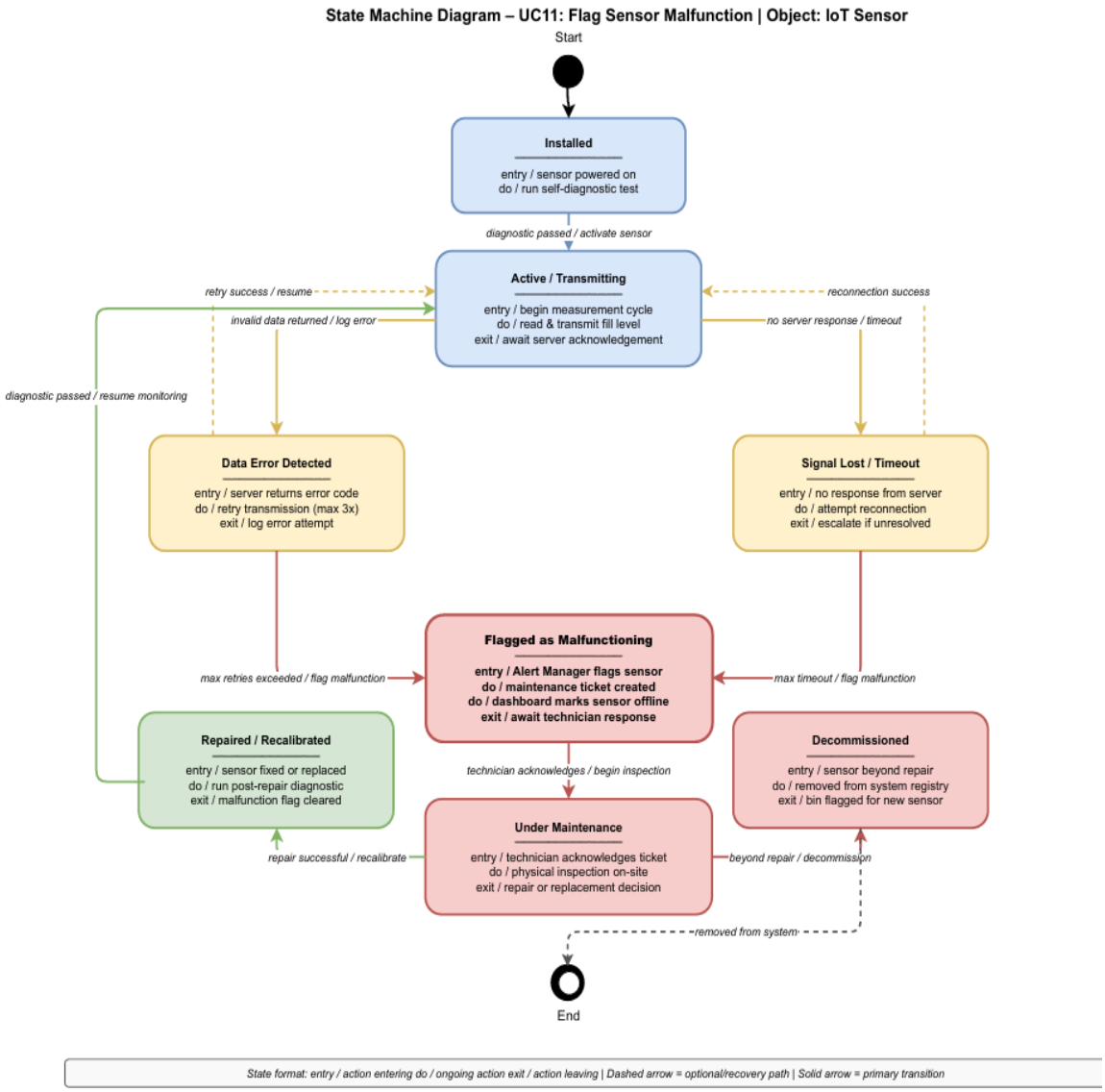


3.4 State Machine Diagram

Use Case 5: Check Fill Threshold

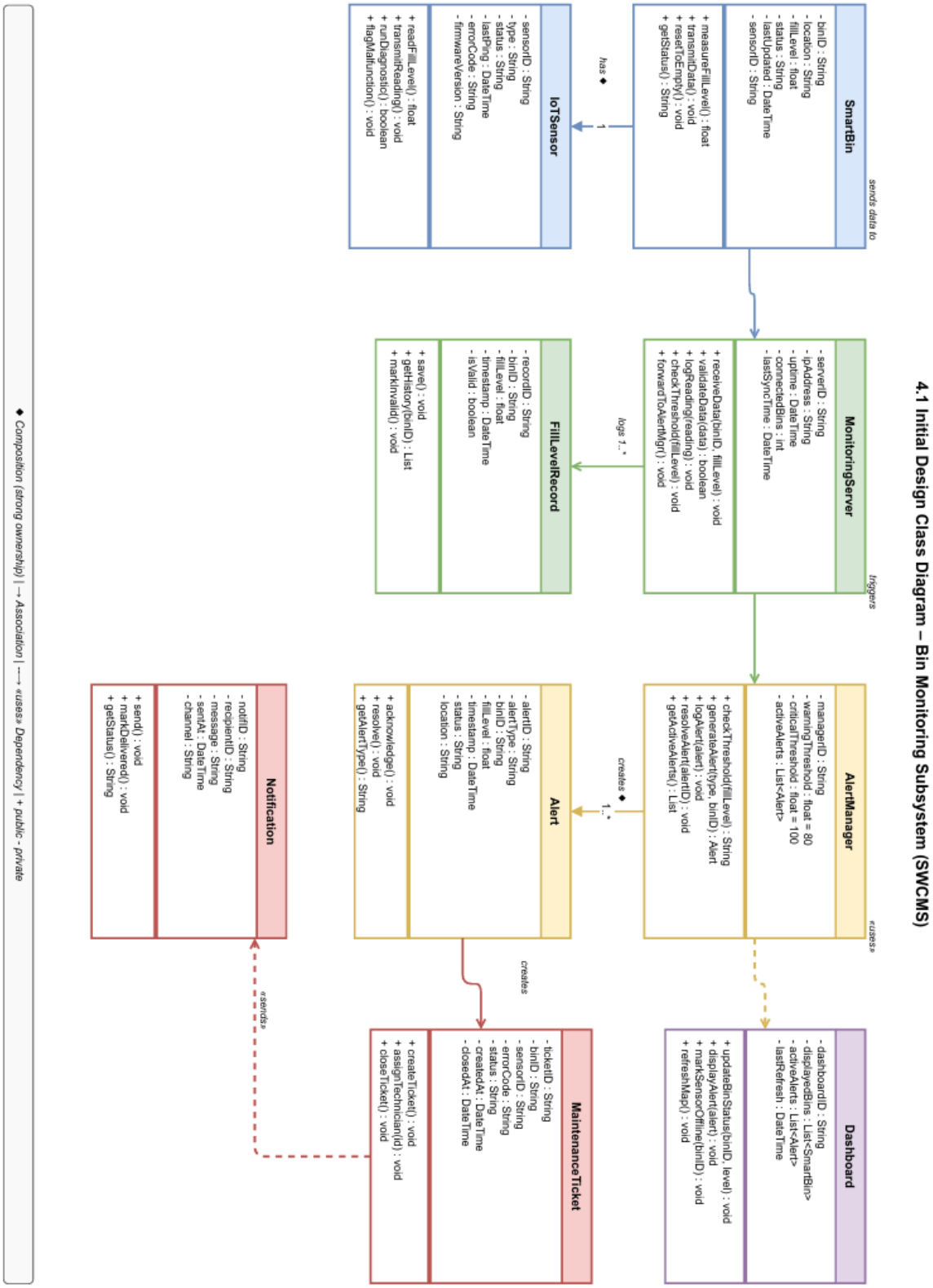


Use Case 11: Flag Sensor Malfunction



4. SYSTEM DESIGN

4.1 Initial Design Class Diagram



4.2 CRC CARDS

SmartBin	
RESPONSIBILITIES	COLLABORATORS
- Store bin location and zone - Track current fill level (%) - Maintain bin status (Active/Full/Offline) - Provide bin ID and metadata	- IoTSensor - FillLevelRecord - Alert - Dashboard

IoTSensor	
RESPONSIBILITIES	COLLABORATORS
- Measure bin fill level via ultrasonic sensing - Transmit data wirelessly to CentralServer - Report its own health/status - Flag itself as malfunctioning if invalid	- SmartBin - CentralServer - FillLevelRecord - MaintenanceReport

FillLevelRecord	
RESPONSIBILITIES	COLLABORATORS
- Log fill-level reading with timestamp - Store binID and sensorID reference - Provide historical data for analytics - Archive or correct old readings	- IoTSensor - CentralServer - Dashboard - Alert

CentralServer	
RESPONSIBILITIES	COLLABORATORS
- Receive and validate incoming sensor data - Log data to FillLevelRecord - Compare fill level to threshold (80%/100%) - Trigger alert generation - Notify Dashboard and RouteOptimizer	- IoTSensor - FillLevelRecord - Alert - Dashboard - RouteOptimizer

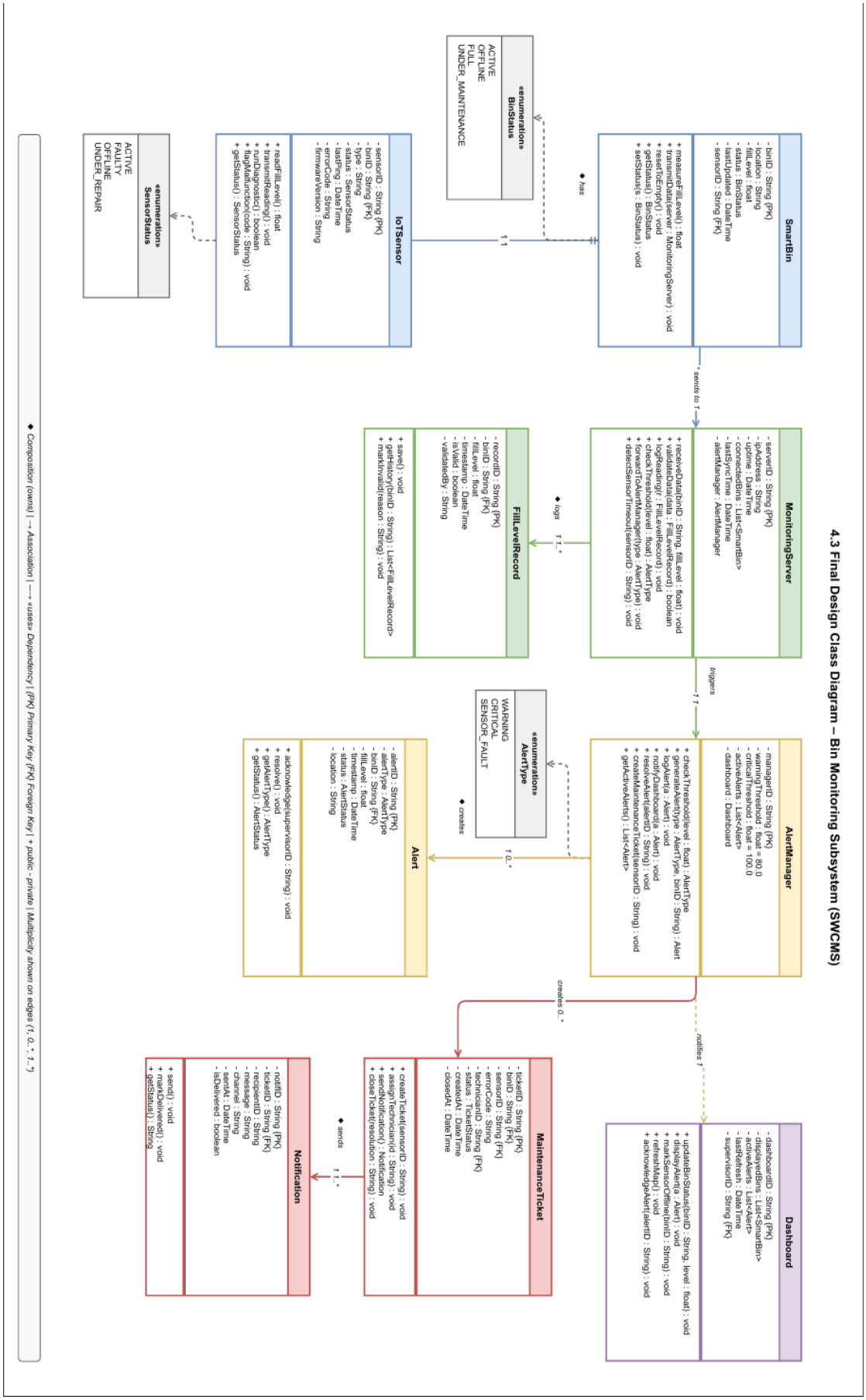
Alert	
RESPONSIBILITIES	COLLABORATORS
- Store alert type (Warning/Critical/Malfunction) - Record timestamp and binID - Track alert status (Active/Resolved) - Support acknowledgement and resolution	- CentralServer - Dashboard - MaintenanceReport - RouteOptimizer

MaintenanceReport	
RESPONSIBILITIES	COLLABORATORS
- Create maintenance ticket for faulty sensor - Track repair status (Open/In-Progress/Closed) - Link to Technician and Sensor - Log repair history	- IoTSensor - Alert - Dashboard - MaintenanceTechnician

Dashboard	
RESPONSIBILITIES	COLLABORATORS
- Display real-time bin fill map - Show active alerts to supervisor - Mark sensors offline when flagged - Refresh status after each data cycle	- CentralServer - Alert - SmartBin - CitySupervisor (actor)

RouteOptimizer	
RESPONSIBILITIES	COLLABORATORS
- Receive critical alert trigger from CentralServer - Calculate optimal truck route for flagged bins - Dispatch route to Fleet Management - Update route when new critical alerts arise	- CentralServer - Alert - SmartBin - FleetManager (external)

4.3 FINAL CLASS DIAGRAM



4.4 SYSTEN SEQUENCE DIAGRAM

